7.4.2 Users and roles

The design principles (DPs) you've been using are provided in appendix B. Let's apply them to design the solution for MPC. DP1 tells you to design before you code, and DP2 and DP3 are about identifying the users and their roles. You can use figure 7.6 to identify the users of the MPC and also the roles of the users. The users of the micropayment channel MPC-Dapp are the organizers who secure the funding for the massive plastic cleanup and the workers who do the recyclable plastics cleanup. Anybody may serve as a worker. In MPC, all you need is the skill to collect plastic garbage and deliver it to an appropriate station. So anybody (with good or bad credit) in the world with an accessible blockchain identity (account address) can join the effort. They don't have to have a bank account. The identities of the organizer and the worker (as well as the smart contract) are their 160-bit account numbers. In this case, these account numbers are the Ethereum blockchain accounts that can be created in a few minutes.

A high school student in Mombasa, Kenya, could generate an identity for themselves, access the MPC web page or app, and initiate opening a microchannel between themselves and an organizer. As they walk every day to school, they collect recyclable plastic garbage, deposit it at a verification station, and get a 0.001 ETH micropayment message from the MPC organizer. On the way back home, they do the same. They collect two bins because they have more time. They get a micropayment of 0.002 ETH after depositing it at a station near their home. After a month or so, their last micropayment is 0.09 ETH. They are happy that they can spend the payment on a movie weekend, and they claim it (0.09 ETH) by sending a message (Tx) to the MPC smart contract. The worker enjoys this